Back

JGR Space Physics

Research Article

Validation and Intercomparison of Ionospheric Electron Density Products From COSMIC-2, FY-3, and Tianmu-1 Radio Occultation Missions

Wenyao Zhang, Binbin Wang, Min Li, Yunbin Yuan, Ting Zhang, Weihong Sun, Chunchun Sheng

First published: 10 October 2026

https://doi.org/10.1029/2025JA034975

VIEW METRICS

EPUB PDF

Get access to the full version of this article. View access options below.

Institutional Login

Access through your institution

Log in with AGU

Log in with your society credentials

Log in

Log in to Wiley Online Library

If you have previously obtained access with your personal account, please log in.

Back

One account for all your research.

Wiley Online Library is part of the CONNECT Network.

Purchase Instant Access

48-Hour online access **$16.00**

Details

Online-only access **$27.00**

Details

PDF download and online access **$62.00**

Details

Check out

Abstract

Multi-source GNSS radio occultation (RO) data are important for space weather monitoring, but instrumental and orbital variations introduce inconsistencies. This study assesses ionospheric data quality across Tianmu-1, FY-3, and COSMIC-2 using over 6.9 million RO events from DOY 122, 2024 to DOY 120, 2025. Validated against ionosondes, NmF2 and hmF2 from all constellations show significant linear correlations (r > 0.8 and 0.6, respectively) with near-normal error distributions; profile validity rates range from 79.88% to 82.94%, decreasing notably in the equatorial ionization anomaly region. In TEC validation against CODE GIMs, all missions exhibited systematic underestimation, with mean differences of –1.04 (FY-3), –3.58 (Tianmu-1), and –3.88 TECU (COSMIC-2), and RMSEs of 8.33, 8.52, and 9.89 TECU, respectively. During this period, the three constellations demonstrate complementary sampling: Tianmu-1 covers a broader spatial range with the

Back

accuracy depends on LEO orbital altitude, with higher-altitude FY-3 showing smaller TEC biases. These results indicate that orbital configuration, sampling distribution, and topside normalization should be jointly considered when integrating multi-source RO data for global ionospheric monitoring.

Plain Language Summary

The ionosphere is an electrically charged layer of Earth's upper atmosphere that can disrupt satellite navigation and communication. To monitor this environment, scientists use low-Earth-orbit satellites that receive signals from navigation constellations. Multiple satellite constellations from different countries—including China's Tianmu-1, the Fengyun-3 (FY-3) series, and the US, COSMIC-2—are collecting vast amounts of global data. However, combining their measurements into a coherent multi-source data set is challenging because they fly at different altitudes and cover different regions. This study analyzes 1 year of continuous global data from these three constellations to identify their unique characteristics. Our findings reveal that their performances are complementary rather than universally superior or inferior. COSMIC-2 provides dense sampling of complex ionospheric structures near the equator, while the polar-orbiting Tianmu-1 contributes critical observations at high latitudes where other constellations have limited reach. Furthermore, satellites flying at higher altitudes, such as FY-3, are less affected by model-dependent extrapolation when estimating the topside ionosphere. These insights provide a practical guide for space weather operators to intelligently combine multi-source satellite data sets, contributing to improved reliability, continuity, and accuracy of global space weather monitoring and warning systems.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Ionosonde data were obtained from the Digital Ionogram DataBase (DIDB) of the Global Ionosphere Radio Observatory (GIRO), which is operated by the University of Massachusetts

Back

archive (https://cddis.nasa.gov/archive/gnss/products/ionex). The Kp index used in this study was obtained from OMNIWeb (https://omniweb.gsfc.nasa.gov/form/dx4.html). The RO data used in this study were obtained from three constellations with differing access policies. COSMIC-2 data, operated by the University Corporation for Atmospheric Research (UCAR), are freely downloadable from the UCAR COSMIC Data Portal (https://data.cosmic.ucar.edu/gnss-ro/cosmic2/rapid/). FY-3 RO data, provided by the National Satellite Meteorological Center (NSMC) of China Meteorological Administration, can be accessed via the NSMC satellite data portal (http://satellite.nsmc.org.cn) after free online registration at https://data.cma.cn/en/#/home. Tianmu-1 RO data are proprietary and subject to access restrictions imposed by the commercial data policy of Aerospace Tianmu (Chongqing) Satellite Technology Co., Ltd., the entity that owns and operates the Tianmu-1 constellation. Under this policy, raw Level-1 and Level-2 RO products are classified as commercial assets and cannot be publicly redistributed without a formal data licensing agreement. Researchers interested in accessing the raw Tianmu-1 data should contact Aerospace Tianmu directly through their official website (https://www.tmsats.com) to inquire about data licensing terms. A limited sample of Level-2 EDP data is available for demonstration at https://www.tmsats.com/openSource. To facilitate reproducibility, and notwithstanding the access restrictions on the raw data, the derived Level-3 ionospheric products used in this study have been deposited in a publicly accessible, registration-free Zenodo repository (https://doi.org/10.5281/zenodo.21966343). These products include the ionospheric parameters (NmF2, hmF2, and VTEC) extracted from the EDPs, as well as the RO–ionosonde matched-pair results used for the reported figures and validation analyses. Together, these derived products allow the main analyses and reported figures to be independently checked without requiring access to the proprietary Tianmu-1 Level-1/Level-2 data.

References

Abdu, M. A., Alam Kherani, E., Batista, I. S., de Paula, E. R., Fritts, D. C., & Sobral, J. H. A. (2009). Gravity wave initiation of equatorial spread F/plasma bubble irregularities based on observational data from the SpreadFEx campaign. *Annales Geophysicae*, **27**(7), 2607–2622. https://doi.org/10.5194/angeo-27-2607-2009

ADS | Web of Science® | Google Scholar

Ai, J., Zhang, X., Li, Y., Sun, Y., Bai, W., Guo, X., et al. (2025). Global thermospheric density response to the May 2024 extreme storm: TianMu-1 constellation observations. *Earth and Planetary Physics*.

Back

Anthes, R. A., Bernhardt, P. A., Chen, Y., Cucurull, L., Dymond, K. F., Ector, D., et al. (2008). The COSMIC/FORMOSAT-3 mission: Early results. *Bulletin of the American Meteorological Society*, **89**(3), 313–334. https://doi.org/10.1175/BAMS-89-3-313

ADS | Web of Science® | Google Scholar

Atabati, A., Jazireeyan, I., Alizadeh, M., Pirooznia, M., Flury, J., Schuh, H., & Soja, B. (2023). Analyzing the ionospheric irregularities caused by the September 2017 geomagnetic storm using ground-based GNSS, swarm, and formosat-3/cosmic data near the equatorial ionization anomaly in East Africa. *Remote Sensing*, **15**(24), 5762. https://doi.org/10.3390/rs15245762

ADS | Web of Science® | Google Scholar

Bartels, J., Heck, N. H., & Johnston, H. F. (1939). The three-hour-range index measuring geomagnetic activity. *Terrestrial Magnetism and Atmospheric Electricity*, **44**(4), 411–454. https://doi.org/10.1029/TE044i004p00411

ADS | Google Scholar

Belehaki, A., Tsagouri, I., Michalareas, G., & Herekakis, T. (2006). F region drift observations from Athens digisonde. *Radio Science*, **41**(6), 1–12. https://doi.org/10.1029/2005RS003322

Web of Science® | Google Scholar

Bianchin, V. B., Marcílio, G. C., Morais, F. P. d., do Ó, R. Q., Barcellos, J. C. E., Marcelino, G. M., & Bezerra, E. A. (2024). *FloripaSat-3: Platform design methodology and preliminar analysis latin american CubeSat workshop (LACW) 2024*. Salvador.

Google Scholar

CASIC. (2023). Tianmu specifications. *China Aerospace Science and Industry Corporation*. Retrieved from https://casichttm.com/en

Google Scholar

Back

variation. *Journal of Geophysical Research: Space Physics*, **128**(10), e2023JA031514. https://doi.org/10.1029/2023JA031514

ADS | Web of Science® | Google Scholar

Cherniak, I., Zakharenkova, I., Braun, J., Wu, Q., Pedatella, N., Schreiner, W., et al. (2021). Accuracy assessment of the quiet-time ionospheric F2 peak parameters as derived from COSMIC-2 multi-GNSS radio occultation measurements. *Journal of Space Weather and Space Climate*, **11**, 18. https://doi.org/10.1051/swsc/2020080

ADS | Web of Science® | Google Scholar

Cherniak, I., Zakharenkova, I., Hunt, D., Weiss, J.-P., Braun, J., Vanhove, T., et al. (2025). Evaluation of commercial GNSS radio occultation ionosphere and space weather data products using COSMIC-2. *GPS Solutions*, **29**(3), 106. https://doi.org/10.1007/s10291-025-01874-0

Web of Science® | Google Scholar

Choudhary, M., Kalita, B. R., Rai, B., Chakrabarty, D., Huy, M. L., Thanh, D. N., et al. (2025). Hemispherically and longitudinally asymmetric ionospheric response to the 23–24 march 2023 geomagnetic disturbances at three pairs of conjugate stations. *Journal of Geophysical Research: Space Physics*, **130**(7), e2024JA033520. https://doi.org/10.1029/2024JA033520

ADS | Web of Science® | Google Scholar

Danzer, J., Scherllin-Pirscher, B., & Foelsche, U. (2013). Systematic residual ionospheric errors in radio occultation data and a potential way to minimize them. *Atmospheric Measurement Techniques*, **6**(8), 2169–2179. https://doi.org/10.5194/amt-6-2169-2013

ADS | Web of Science® | Google Scholar

Dietrich, N., Matsuo, T., Lin, C.-Y., diLorenzo, B., Chien-Hung Lin, C., & Fang, T.-W. (2024). Evaluating radio occultation (RO) constellation designs using observing system simulation experiments (OSSEs) for ionospheric specification. *Space Weather*, **22**(8), e2024SW003958. https://doi.org/10.1029/2024SW003958

Back

Donaldson, J. E., Parker, J. J. K., Moreau, M. C., Highsmith, D. E., & Martzen, P. D. (2020). Characterization of on-orbit GPS transmit antenna patterns for space users. *Navigation*, **67**(2), 411–438. https://doi.org/10.1002/navi.361

ADS | Web of Science® | Google Scholar

EUMETSAT. (2025). Sentinel-6 A GNSS-RO NTC cal/val report (p. v1G).

Google Scholar

Friis, H. T. (1946). A note on a simple transmission formula. *Proceedings of the IRE*, **34**(5), 254–256. https://doi.org/10.1109/JRPROC.1946.234568

ADS | Web of Science® | Google Scholar

Fu, N., Zhang, R., Pan, Q., Tian, X., Long, S., & Yao, C. (2023). Quality assessment of the ionospheric density profiles based on long-term COSMIC 1 and 2 radio occultation observations. *Journal of Earth System Science*, **132**(4), 175. https://doi.org/10.1007/s12040-023-02190-2

ADS | Web of Science® | Google Scholar

Galkin, I. A., & Reinisch, B. W. (2008). The new ARTIST 5 for all digisondes. *Ionosonde Network Advisory Group Bulletin*, **69**(8), 1–8.

Google Scholar

Galkin, I. A., Reinisch, B. W., Huang, X., & Khmyrov, G. M. (2013). Confidence score of ARTIST-5 ionogram autoscaling. *Ionosonde Network Advisory Group (INAG) Bulletin*, **73**, 1–7.

Google Scholar

Gorbunov, M., Kirchengast, G., & Lauritsen, K. B. (2021). Generalized canonical transform method for radio occultation sounding with improved retrieval in the presence of horizontal gradients. *Atmospheric Measurement Techniques*, **14**(2), 853–867. https://doi.org/10.5194/amt-14-853-2021

Back

Gorbunov, M. E., & Kornblueh, L. (2001). Analysis and validation of GPS/MET radio occultation data. *Journal of Geophysical Research*, **106**(D15), 17161–17169. https://doi.org/10.1029/2000jd900816

ADS | Web of Science® | Google Scholar

Guo, P., Xu, X., & Zhang, G. X. (2011). Analysis of the ionospheric equivalent slab thickness based on ground-based GPS/TEC and GPS/COSMIC RO measurements. *Journal of Atmospheric and Solar-Terrestrial Physics*, **73**(7), 839–846. https://doi.org/10.1016/j.jastp.2011.02.002

ADS | Web of Science® | Google Scholar

Habarulema, J. B., Katamzi, Z. T., & Yizengaw, E. (2014). A simultaneous study of ionospheric parameters derived from FORMOSAT-3/COSMIC, GRACE, and CHAMP missions over middle, low, and equatorial latitudes: Comparison with ionosonde data. *Journal of Geophysical Research: Space Physics*, **119**(9), 7732–7744. https://doi.org/10.1002/2014JA020192

ADS | Web of Science® | Google Scholar

He, Y., Zhang, S., Guo, S., & Wu, Y. (2023). Quality assessment of the atmospheric radio occultation profiles from FY-3E/GNOS-II BDS and GPS measurements. *Remote Sensing*, **15**(22), 5313. https://doi.org/10.3390/rs15225313

ADS | Web of Science® | Google Scholar

Ho, S.-p., Zhou, X., Shao, X., Chen, Y., Jing, X., & Miller, W. (2023). Using the commercial GNSS RO spire data in the neutral atmosphere for climate and weather prediction studies. *Remote Sensing*, **15**(19), 4836. https://doi.org/10.3390/rs15194836

ADS | Web of Science® | Google Scholar

Hocke, K. (1997). Inversion of GPS meteorology data. *Annales Geophysicae*, **15**, 443–450. https://doi.org/10.1007/s00585-997-0443-1

ADS | Google Scholar

Back

Observations and Remote Sensing, **18**, 16858–16867. https://doi.org/10.1109/JSTARS.2025.3582010

ADS | Web of Science® | Google Scholar

Hu, T., Xu, X., & Luo, J. (2024). Multi-source data ingestion for IRI-2020 model: A combination of ground-based and space-borne observations. *GPS Solutions*, **28**(2), 78. https://doi.org/10.1007/s10291-024-01620-y

Web of Science® | Google Scholar

Huang, B., Chen, P., Yao, Y., Wang, R., & Zhang, Y. (2024). Validation of ionospheric parameters from electron density profiles of FY-3E satellite using ionosonde, GIMs, satellite altimetry and other occultations. *Journal of Geophysical Research: Space Physics*, **129**(12), e2024JA032590. https://doi.org/10.1029/2024ja032590

CAS | ADS | Web of Science® | Google Scholar

Huang, F., Yin, C., Liu, Y., Sun, Y., Xia, J., Bai, W., et al. (2025). Tianmu-1 constellation GNSS-R In-orbit performance: Spatiotemporal characteristics, product applications and polarimetric features. *IEEE Transactions on Geoscience and Remote Sensing*, **63**, 1–20. https://doi.org/10.1109/TGRS.2025.3581343

PubMed | Web of Science® | Google Scholar

Jakowski, N. (2005). Radio occultation techniques for probing the ionosphere. *URSI Radio Science Bulletin*, **2005**(314), 4–15. https://doi.org/10.23919/URSIRSB.2005.7909740

Google Scholar

Jiang, C., An, Q., Wang, S., Nie, W., Zhu, H., & Liu, G. (2024). Accuracy assessment of the ionospheric total electron content derived from COSMIC-2 radio occultation based on multi-source data. *Advances in Space Research*, **73**(10), 5157–5170. https://doi.org/10.1016/j.asr.2024.02.051

CAS | ADS | Web of Science® | Google Scholar

Lee, W., Liu, G., Wu, D. L., & Rowland, D. E. (2025). Ionospheric response to the 10 May 2024 geomagnetic storm as observed in GNSS radio occultation electron density. *Journal of Geophysical*

Back

ADS | Web of Science® | Google Scholar

Lei, J., Syndergaard, S., Burns, A. G., Solomon, S. C., Wang, W., Zeng, Z., et al. (2007). Comparison of COSMIC ionospheric measurements with ground-based observations and model predictions: Preliminary results. *Journal of Geophysical Research*, **112**(A7). https://doi.org/10.1029/2006JA012240

Web of Science® | Google Scholar

Li, M., & Yue, X. (2021). Statistically analyzing the effect of ionospheric irregularity on GNSS radio occultation atmospheric measurement. *Atmospheric Measurement Techniques*, **14**(4), 3003–3013. https://doi.org/10.5194/amt-14-3003-2021

ADS | Web of Science® | Google Scholar

Liao, M., Zhang, P., Bi, Y., & Yang, G. (2015a). A preliminary estimation of the radio occultation products accuracy from the Fengyun-3C meteorological satellite. *Acta Meteorologica Sinica*, **73**(6), 1131–1140. https://doi.org/10.11676/qxxb2015.072

ADS | Google Scholar

Liao, M., Zhang, P., Bi, Y., & Yang, G. (2015b). A preliminary estimation of the radio occultation products accuracy from the Fengyun-3C meteorological satellite. *Acta Meteorol*, **73**(6), 1131–1140. https://doi.org/10.11676/qxxb2015.072

ADS | Google Scholar

Lin, C.-Y., Lin, C. C.-H., Liu, J.-Y., Rajesh, P. K., Matsuo, T., Chou, M.-Y., et al. (2020). The early results and validation of FORMOSAT-7/COSMIC-2 space weather products: Global ionospheric specification and Ne-Aided abel electron density profile. *Journal of Geophysical Research: Space Physics*, **125**(10), e2020JA028028. https://doi.org/10.1029/2020JA028028

ADS | Web of Science® | Google Scholar

Liu, H., Huo, X., Zhang, T., Yuan, Y., & Sun, W. (2025a). Enhancing the BDGIM model for ionospheric delay correction using GNSS LEO occultation data during high solar activity. *GPS Solutions*, **29**(3), 147. https://doi.org/10.1007/s10291-025-01908-7

Back

Liu, X., Bu, J., Zuo, X., Wang, Z., Wang, Q., Wang, Q., et al. (2025b). Performance validation of sea surface wind speed retrieval algorithms and products from the Chinese Tianmu-1 constellation GNSS-R: First results on comparison with other wind speed products. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, **18**, 5189–5203. https://doi.org/10.1109/JSTARS.2025.3527026

ADS | Web of Science® | Google Scholar

Luntama, J.-P., Kirchengast, G., Borsche, M., Foelsche, U., Steiner, A., Healy, S., et al. (2008). Prospects of the EPS GRAS mission for operational atmospheric applications. *Bulletin of the American Meteorological Society*, **89**(12), 1863–1876. https://doi.org/10.1175/2008BAMS2399.1

ADS | Web of Science® | Google Scholar

Luo, J., Wang, H., Xu, X., & Sun, F. (2019). The influence of the spatial and temporal collocation windows on the comparisons of the ionospheric characteristic parameters derived from COSMIC radio occultation and digisondes. *Advances in Space Research*, **63**(10), 3088–3101. https://doi.org/10.1016/j.asr.2019.01.024

ADS | Web of Science® | Google Scholar

Mannucci, A. J., Ao, C. O., & Williamson, W. (2020). GNSS radio occultation. In *Position, navigation, and timing technologies in the 21st century* (pp. 971–1013). https://doi.org/10.1002/9781119458449.ch33

Google Scholar

Matzka, J., Stolle, C., Yamazaki, Y., Bronkalla, O., & Morschhauser, A. (2021). The geomagnetic Kp index and derived indices of geomagnetic activity. *Space Weather*, **19**, e2020SW002641. https://doi.org/10.1029/2020SW002641

ADS | Web of Science® | Google Scholar

Maute, A., & Richmond, A. D. (2017). F-Region dynamo simulations at low and mid-latitude. *Space Science Reviews*, **206**(1–4), 471–493. https://doi.org/10.1007/s11214-016-0262-3

Back

Mo, Z., Lou, Y., Zhang, W., Zhou, Y., Wu, P., & Zhang, Z. (2024). Performance assessment of multi-source GNSS radio occultation from COSMIC-2, MetOp-B/C, FY-3D/E, spire and PlanetiQ over China. *Atmospheric Research*, **311**, 107704. https://doi.org/10.1016/j.atmosres.2024.107704

CAS | Web of Science® | Google Scholar

Nickolaenko, A. P. (2009). Concept of planetary thermal balance and global warming. *Journal of Geophysical Research*, **114**(A4). https://doi.org/10.1029/2008JA013753

Google Scholar

Nishioka, M., Saito, A., & Tsugawa, T. (2008). Occurrence characteristics of plasma bubble derived from global ground-based GPS receiver networks. *Journal of Geophysical Research*, **113**(A5). https://doi.org/10.1029/2007JA012605

Web of Science® | Google Scholar

Niu, J., Fang, H., & Weng, L. (2014). Establishment of NmF2 empirical model in the northern hemisphere based on COSMIC occultation data. *Science China Technological Sciences*, **57**(2), 339–344. https://doi.org/10.1007/s11431-013-5447-3

ADS | Web of Science® | Google Scholar

NSMC. (2021). Fengyun-3 series satellite overview and technical specifications. *National Satellite Meteorological Center of CMA*. Retrieved from http://www.nsmc.org.cn/nsmc/cn/satellite/FY3.html

Google Scholar

Paul, K., Haralambous, H., Moses, M., & Panda, S. (2025). Global accuracy assessment of ionospheric F2 peak characteristics based on coincident-colocated COSMIC-2 RO and digisonde measurements: A three-year period analysis (2020-2022). *Advances in Space Research*, **77**(1), 1166–1185. https://doi.org/10.1016/j.asr.2025.09.083

ADS | Web of Science® | Google Scholar

Back

Retrieved from https://apps.dtic.mil/sti/html/trecms/AD1174448/

Google Scholar

Ren, X., Zhang, J., Chen, J., & Zhang, X. (2022). Global ionospheric modeling using Multi-GNSS and upcoming LEO constellations: Two methods and comparison. *IEEE Transactions on Geoscience and Remote Sensing*, **60**, 1–15. https://doi.org/10.1109/TGRS.2021.3050413

PubMed | Web of Science® | Google Scholar

Rocken, C., Anthes, R., Exner, M., Hunt, D., Sokolovskiy, S., Ware, R., et al. (1997). Analysis and validation of GPS/MET data in the neutral atmosphere. *Journal of Geophysical Research*, **102**(D25), 29849–29866. https://doi.org/10.1029/97JD02400

ADS | Web of Science® | Google Scholar

Schaer, S., Beutler, G., Rothacher, M., & Springer, T. A. (1996). Daily global ionosphere maps based on GPS carrier phase data routinely produced by the CODE analysis center. In *Proceedings of the IGS analysis center workshop 1996*.

Google Scholar

Schaer, S., & géodésique, S. h. d. s. n. C. (1999). *Mapping and predicting the earth's ionosphere using the global positioning system* (Vol. **59**). Institut für Geodäsie und Photogrammetrie, Eidg. Technische Hochschule.

Google Scholar

Schreiner, W., Rocken, C., Sokolovskiy, S., Syndergaard, S., & Hunt, D. (2007). Estimates of the precision of GPS radio occultations from the COSMIC/FORMOSAT-3 mission. *Geophysical Research Letters*, **34**(4). https://doi.org/10.1029/2006GL027557

Web of Science® | Google Scholar

Schreiner, W. S., Sokolovskiy, S. V., Rocken, C., & Hunt, D. C. (1999). Analysis and validation of GPS/MET radio occultation data in the ionosphere. *Radio Science*, **34**(4), 949–966. https://doi.org/10.1

Back

ADS | Web of Science® | Google Scholar

Schreiner, W. S., Weiss, J. P., Anthes, R. A., Braun, J., Chu, V., Fong, J., et al. (2020). COSMIC-2 radio occultation constellation: First results. *Geophysical Research Letters*, **47**(4), e2019GL086841. https://doi.org/10.1029/2019GL086841

ADS | Web of Science® | Google Scholar

Shi, S., Li, W., Zhang, K., Wu, S., Shi, J., Song, F., & Sun, P. (2021). Validation of COSMIC-2-Derived ionospheric peak parameters using measurements of ionosondes. *Remote Sensing*, **13**(21), 4238. https://doi.org/10.3390/rs13214238

ADS | Web of Science® | Google Scholar

Singh, R., Ojha, S. P., Anthes, R., & Hunt, D. (2021). Evaluation and assimilation of the COSMIC-2 radio occultation constellation observed atmospheric refractivity in the WRF data assimilation system. *Journal of Geophysical Research: Atmospheres*, **126**(18), e2021JD034935. https://doi.org/10.1029/2021JD034935

ADS | Web of Science® | Google Scholar

Su, F., & Wang, W. (2022). Diurnal and seasonal characteristics of the longitudinal variations of electron densities in the topside ionosphere at middle latitudes. *Journal of Geophysical Research: Space Physics*, **127**(6), e2022JA030308. https://doi.org/10.1029/2022JA030308

ADS | Web of Science® | Google Scholar

Sun, Y., Huang, F., Xia, J., Yin, C., Bai, W., Du, Q., et al. (2023). GNOS-II on Fengyun-3 satellite series: Exploration of Multi-GNSS reflection signals for operational applications. *Remote Sensing*, **15**(24), 5756. https://doi.org/10.3390/rs15245756

ADS | Web of Science® | Google Scholar

Swarnalingam, N., Wu, D. L., Emmons, D. J., & Gardiner-Garden, R. (2023). Optimal estimation inversion of ionospheric electron density from GNSS-POD limb measurements: Part II-Validation and

Back

ADS | Web of Science® | Google Scholar

Tu, J., He, X., Xu, X., Song, M., & Xu, X. (2025). Assessment of Tianmu-1 multi-GNSS-R global soil moisture products. *Advances in Space Research*, **76**(3), 1476–1491. https://doi.org/10.1016/j.asr.2025.05.066

ADS | Web of Science® | Google Scholar

Wang, H., Luo, J., & Xu, X. (2019). Ionospheric peak parameters retrieved from FY-3C radio occultation: A statistical comparison with measurements from COSMIC RO and digisondes over the globe. *Remote Sensing*, **11**(12), 1419. https://doi.org/10.3390/rs11121419

ADS | Web of Science® | Google Scholar

Ware, R., Exner, M., Feng, D., Gorbunov, M., Hardy, K., Herman, B., et al. (1996). GPS sounding of the atmosphere from low Earth orbit: Preliminary results. *Bulletin of the American Meteorological Society*, **77**(1), 19–40. https://doi.org/10.1175/1520-0477(1996)077<0019:GSOTAF>2.0.CO;2

ADS | Web of Science® | Google Scholar

Weiss, J.-P., Schreiner, W. S., Braun, J. J., Xia-Serafino, W., & Huang, C.-Y. (2022). COSMIC-2 mission summary at three years in orbit. *Atmosphere*, **13**(9), 1409. https://doi.org/10.3390/atmos13091409

ADS | Web of Science® | Google Scholar

Wickert, J., Beyerle, G., König, R., Heise, S., Grunwaldt, L., Michalak, G., et al. (2005). GPS radio occultation with CHAMP and GRACE: A first look at a new and promising satellite configuration for global atmospheric sounding. *Annales Geophysicae*, **23**, 653–658. https://doi.org/10.5194/angeo-23-653-2005

ADS | Web of Science® | Google Scholar

Wu, D. L., Emmons, D. J., & Swarnalingam, N. (2022). Global GNSS-RO electron density in the lower ionosphere. *Remote Sensing*, **14**(7), 1577. https://doi.org/10.3390/rs14071577

Back

Xia, P., Ye, S., Jiang, K., & Chen, D. (2017). Estimation and evaluation of COSMIC radio occultation excess phase using undifferenced measurements. *Atmospheric Measurement Techniques*, **10**(5), 1813–1821. https://doi.org/10.5194/amt-10-1813-2017

ADS | Web of Science® | Google Scholar

Xu, J., Wang, Q., Ke, F., Zhang, B., Yan, C., & Zhang, H. (2021). Analysis of ionospheric parameters retrieved from feng-yun 3C and COSMIC radio occultation. *Advances in Space Research*, **68**(1), 214–224. https://doi.org/10.1016/j.asr.2021.03.005

ADS | Web of Science® | Google Scholar

Yang, D., & Fang, H. (2023). A low-latitude three-dimensional ionospheric electron density model based on radio occultation data using artificial neural networks with prior knowledge. *Space Weather*, **21**(1), e2022SW003299. https://doi.org/10.1029/2022SW003299

ADS | Web of Science® | Google Scholar

Yang, J., Huang, W., Xia, G., Zhou, C., & Chen, Y. (2024). Operational forecasting of global ionospheric TEC maps 1-2-and 3-day in advance by CONVLSTM model. *Remote Sensing*, **16**(10), 1700. https://doi.org/10.3390/rs16101700

ADS | Web of Science® | Google Scholar

Yang, K.-F., Chu, Y.-H., Su, C.-L., Ko, H.-T., & Wang, C.-Y. (2009). An examination of FORMOSAT-3/COSMIC ionospheric electron density profile: Data quality criteria and comparisons with the IRI model. *TAO: Terrestrial, Atmospheric and Oceanic Sciences*, **20**(1), 1. https://doi.org/10.3319/TAO.2007.10.05.01(F3C

ADS | Web of Science® | Google Scholar

Yang, W., Guo, F., Zhang, X., Zhu, Y., Li, Z., & Zhang, Z. (2025). First quasi-global soil moisture retrieval using Fengyun-3 GNSS-R constellation observations. *Remote Sensing of Environment*, **321**, 114653. https://doi.org/10.1016/j.rse.2025.114653

Web of Science® | Google Scholar

Back

conference.

Google Scholar

Yin, Y., González-Casado, G., Rovira-Garcia, A., Juan, J. M., Sanz, J., & Shao, Y. (2022). Summer nighttime anomalies of ionospheric electron content at midlatitudes: Comparing years of low and high solar activities using observations and tidal/planetary wave features. *Remote Sensing*, **14**(5), 1237. https://doi.org/10.3390/rs14051237

ADS | Web of Science® | Google Scholar

Yue, X., Schreiner, W. S., Pedatella, N., Anthes, R. A., Mannucci, A. J., Straus, P. R., & Liu, J.-Y. (2014). Space weather observations by GNSS radio occultation: From FORMOSAT-3/COSMIC to FORMOSAT-7/COSMIC-2. *Space Weather*, **12**(11), 616–621. https://doi.org/10.1002/2014SW001133

ADS | PubMed | Web of Science® | Google Scholar

Zhu, S., Li, D., Gao, S., Xie, Q., & Pan, X. (2025). Improving "23.07" heavy rainstorm simulation through assimilating FY-3E MWTS-3 radiance data. *Earth and Space Science*, **12**(5), e2025EA004363. https://doi.org/10.1029/2025EA004363

ADS | Web of Science® | Google Scholar

Zou, L., Rishbeth, H., Müller-Wodarg, I. C. F., Aylward, A. D., Millward, G. H., Fuller-Rowell, T. J., et al. (2000). Annual and semiannual variations in the ionospheric F2-layer. I. Modelling. *Annales Geophysicae*, **18**(8), 927–944. https://doi.org/10.1007/s00585-000-0927-8

CAS | ADS | Web of Science® | Google Scholar

Download PDF

Back to Top

Back

AGU PUBLICATIONS

AGU.ORG

AGU MEMBERSHIP

RESOURCES

PUBLICATION INFO

© 2026 American Geophysical Union

ABOUT WILEY ONLINE LIBRARY

Privacy Policy
Terms of Use
About Cookies
Manage Cookies
Accessibility
Wiley Research DE&I Statement and Publishing Policies

HELP & SUPPORT

Contact Us
Training and Support
DMCA & Reporting Piracy
Sitemap

OPPORTUNITIES

Subscription Agents
Advertisers & Corporate Partners

CONNECT WITH WILEY

The Wiley Network
Wiley Press Room

Back

and training of artificial intelligence technologies or similar technologies.